Jafar Banar

Senior Data & AI Engineer | Python · LLM · ML · 5G/6G | PhD candidate, Telecom & Radar (Chalmers)

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PROFESSIONAL SUMMARY

Senior Data Engineer and AI/ML engineer with 10+ years across automotive data systems, AI/ML, radar/perception, and telecom. Senior Data Engineer consultant at Sigma Embedded Engineering Yora AB, most recently on assignment at Volvo Cars building scalable test-data platforms, ETL/ELT pipelines, and production-grade data engineering on Azure. Previously optimized an on-prem LLM/RAG platform (response time from 5–8s to 2–4s, memory from 12 GB to 8 GB) and validated automotive radar/LiDAR perception against ground truth. Dual-degree PhD candidate (Chalmers / TU Eindhoven) in synchronization, distributed 5G/6G, and coherent radar. Independent work on agentic AI with verifiable output: citation-grounded retrieval, schema-validated responses and deterministic fallbacks.

CORE SKILLS

- **AI/ML & Data Engineering:** LLM/RAG, BM25/lexical retrieval, agentic systems, structured-output validation, LLM evaluation, NLP, embeddings, PyTorch, XGBoost, scikit-learn, ONNX quantization (INT8/FP16), ETL/ELT pipelines, automotive test data, large-scale data processing, data quality, Grafana
- **Backend & Cloud:** Python, SQL, FastAPI, Redis, Celery, PostgreSQL, Qdrant, FAISS/ChromaDB, Docker, Kubernetes, CI/CD, Linux, Git, Azure, AWS
- **Automotive / Radar / Embedded:** Automotive radar, ADAS, sensor fusion, C/C++, Go, ROS, LCM, OpenCV, CAN, ISO 26262, SOTIF, FMCW radar, bistatic radar
- **Telecom / Wireless:** 4G/5G/6G, 3GPP, OFDM, distributed MIMO, beamforming, precoding, SDR/USRP, KPI/QoS analysis, synchronization, calibration
- **Languages & Licence:** English (Fluent), Swedish (Intermediate, SFI C completed) · Driving licence: Swedish B

PROFESSIONAL EXPERIENCE

Senior Data Engineer (Consultant) May 2026 – Present
Sigma Embedded Engineering Yora AB, *Gothenburg, Sweden*

- Client assignment at **Volvo Cars** as **Senior Data Engineer** (May 2026 to Sep 2026), engineering scalable automotive test-data platforms, ETL/ELT pipelines, and production-grade data workflows.
- Developed Python/SQL data-processing workflows for large-scale vehicle/test-bed data; delivered QA-validated production changes that made measurement-data (INCA/TAS) sync-candidate selection deterministic and stabilized QA-versus-production data comparisons.
- Supported CI/CD, logging, monitoring, and operational reliability; reviewed integration changes spanning Azure storage routing, data extraction, and CIFS/NAS sources.

Tech: *Python, SQL, ASAM MDF4, Sympathy for Data, dSpace HIL, Azure/AKS, GitHub Actions, Grafana, Oracle, MongoDB, CIFS/NAS*

Senior Development Engineer (AI Engineer) Feb 2025 – Jul 2026
QRTech AB, *Gothenburg, Sweden*

- Optimized an on-prem LLM/RAG platform by profiling retrieval, caching, and inference, cutting response time from 5–8s to 2–4s and memory from 12 GB to 8 GB.
- Co-developed, as part of a team, a LangGraph test-specification generator spanning 7 standards, 3 markets, 5 artifact formats, and 8 API endpoints; wrote the system requirement specification, interface control document and verification plan for an agricultural robot MVP, contributing to its hazard analysis and risk assessment (HARA).
- Delivered document anonymization with PII detection and AI CV-matching workflows across multiple document formats.
- Client assignment at **Einride** as **Senior Perception Engineer** (Sep 2025 to Dec 2025), via consultancy. Found and fixed a Go perception bug that under-reported target speed by $\cos^2$(elevation) (3.0 m/s read as 0.75 m/s at 60°), merged after validation on 22 scenarios (~360k radar clusters); proposed a hybrid RANSAC, weighted-SVD and Kalman estimator.
- Einride: built replay and KPI workflows against OXTS ground truth, analyzed LiDAR timing over 1.86M+ packets, verified LiDAR detection limits against requirements, and contributed to a SOTIF LiDAR+Radar feasibility prototype.

Tech: *Python, FastAPI, Redis, Celery, LangGraph, FAISS/ChromaDB, Qdrant, PostgreSQL, spaCy, Docker/Kubernetes, GitHub Actions, Jenkins; Einride assignment: Go, C++, Python, ROS1, LCM, PyTorch, OpenCV, OXTS, MCAP/Foxglove, SOTIF*

Founder, CEO & AI Engineer (part-time) Jul 2024 – Present
AIOpt AB, *Gothenburg, Sweden*

- Built a safety and evidence layer for AI-driven changes to 4G/5G networks: a seven-gate deterministic decision engine that approves, escalates or refuses each proposed change, checked against the official 3GPP TS 28.532 OpenAPI by its own tests and run end-to-end on a real 5G software stack in the lab (~9 ms mean decision latency).

- Designed a tamper-evident evidence format (append-only, hash-chained, Ed25519 countersigned, publicly timestamped) with an MIT-licensed offline verifier that shares no code with the writer; KPI forecasting, anomaly detection and root-cause analysis in PyTorch/XGBoost. Almi verification funding (50k SEK), EUIPO SME Fund, NVIDIA Inception.

Visiting Researcher, Automotive Radar (PhD secondment) Jun 2021 – Nov 2022
NXP Semiconductors, *Eindhoven, Netherlands*

- Seconded from Chalmers (EU MyWave), developing synchronization and signal-processing algorithms for coherent distributed automotive radar used in ADAS.
- Simulated, validated, and prepared radar algorithm models for embedded deployment in collaboration with hardware and software teams.

Tech: *MATLAB, C++, Python, LabVIEW, signal processing, automotive radar, embedded deployment*

Doctoral Researcher & Project Assistant (Wireless Systems) Jul 2020 – Jan 2025
Chalmers University of Technology, *Gothenburg, Sweden*

- Doctoral researcher Jul 2020 to Dec 2024 (dual degree with TU Eindhoven, incl. NXP placement), then Project Assistant in Jan 2025 on distributed MIMO synchronization; defence February 2027.
- Research distributed massive-MIMO synchronization, OFDM impairment modeling, and coherent radar; built MATLAB/Python simulation frameworks and ran USRP-based experiments. Supervised 10+ thesis students (7 MSc theses, 2 bachelor projects incl. FMCW radar with SDR) with Ericsson, Qamcom, KTH and others.

Tech: *MATLAB, Python, SDR/USRP, distributed MIMO, OFDM, synchronization, calibration, beamforming, precoding*

RF Optimization Engineer Nov 2019 – Jun 2020
AEC - RavanErtebat (Access Endless Communication)

- Optimized a live Ericsson 2G/3G/4G-LTE network for MTN: daily KPI reporting, parameter-tuning trials incl. a power-offset trial to cut dropped calls, NodeB audits and trouble-ticket root-cause analysis.

Tech: *Ericsson RAN (RNC/NodeB), TEMS Discovery, Linux, Bash, KPI analysis, RF planning, drive testing*

Telecom Researcher Apr 2016 – Nov 2019
ICT Research Institute

- Research and simulation studies on 5G, IoT, big data and cloud computing, with technical reports and trainings on new telecom technologies.

Tech: *MATLAB, Python, C++, telecom systems analysis, 3GPP/ETSI standards, simulation*

SELECTED PROJECTS

- Published groundcite on PyPI: a zero-dependency Python library and CLI that verifies an LLM answer cites only the sources it was supplied, measures citation coverage per factual sentence, and fails a CI build on violation. Extracted from the 3GPP assistant below, which now depends on it. Tested on Python 3.9-3.13 (pypi.org/project/groundcite).
- Built a citation-grounded retrieval assistant over 3GPP 5G NR specifications with a from-scratch BM25 index and programmatic citation validation that rejects fabricated clause references, reaching MRR 0.957 and hit-rate@1 of 0.913 on a 23-question hand-labelled evaluation set.
- Built three production-oriented AI agents (inbox triage with human approval, invoice extraction with deterministic anomaly checks, uptime monitoring with LLM incident analysis), each with schema-validated output and rule-based fallbacks; and an ONNX edge-deployment pipeline (INT8/FP16 quantization, p50/p95 benchmarks, CI regression gate, ARM64/Jetson).

EDUCATION

PhD, Electrical Engineering (Dual Degree) Jul 2020 – Feb 2027 (expected)
Chalmers University of Technology / TU Eindhoven, *Sweden / Netherlands*

- Thesis: digital array calibration and synchronization for distributed massive MIMO and distributed radars. University employment ended Jan 2025; defence scheduled February 2027.

MSc, Electrical Engineering 2014 – 2016
IUST

- Thesis: uplink data rate improvement in full-duplex cooperative OFDMA networks.

BSc, Electrical Engineering 2009 – 2014
AUT

- Thesis: data transmission protocols between ground stations and LEO satellites.
- Training: OxML 2024 Oxford Machine Learning Summer School, AI for Global Goals (generative AI), MLx Fundamentals 2024, BSE LLM course, Applied Project Management (Chalmers), PhD Project Planning (TU/e).
- 3rd Place, IEEE ISIT 2025 “Wireless Wanderlust” (D2I) challenge, deep learning for CSI-based localization.